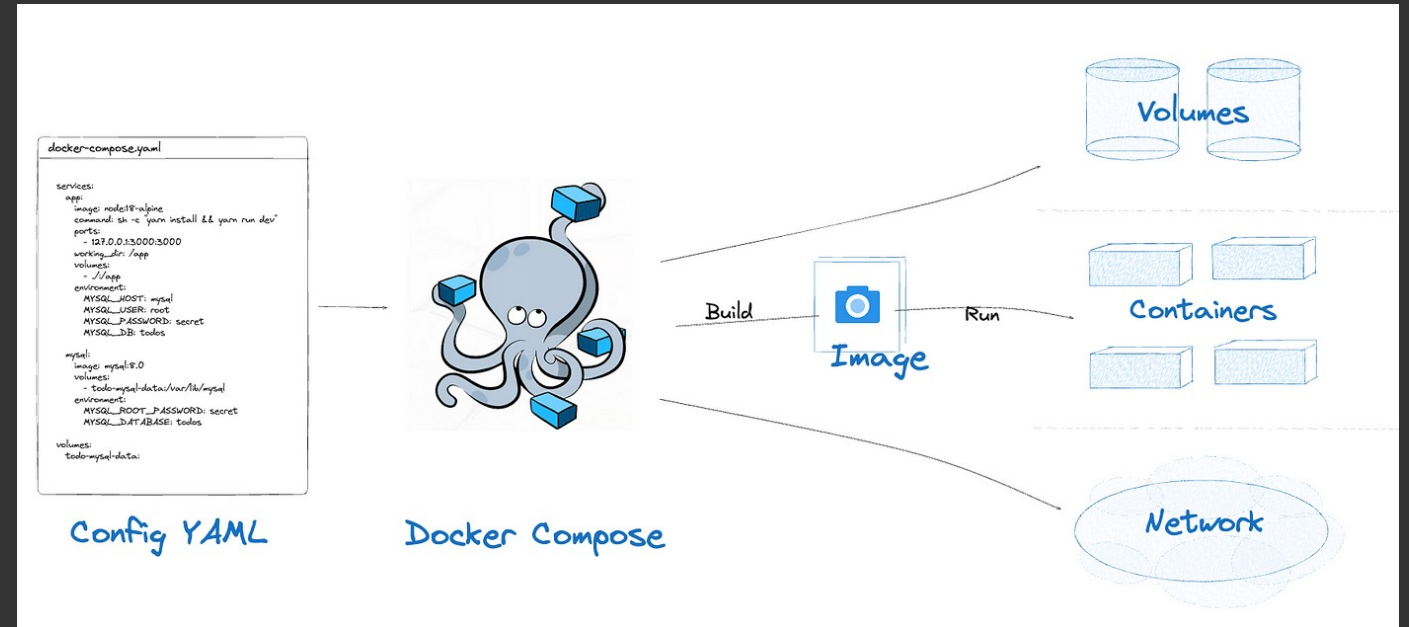
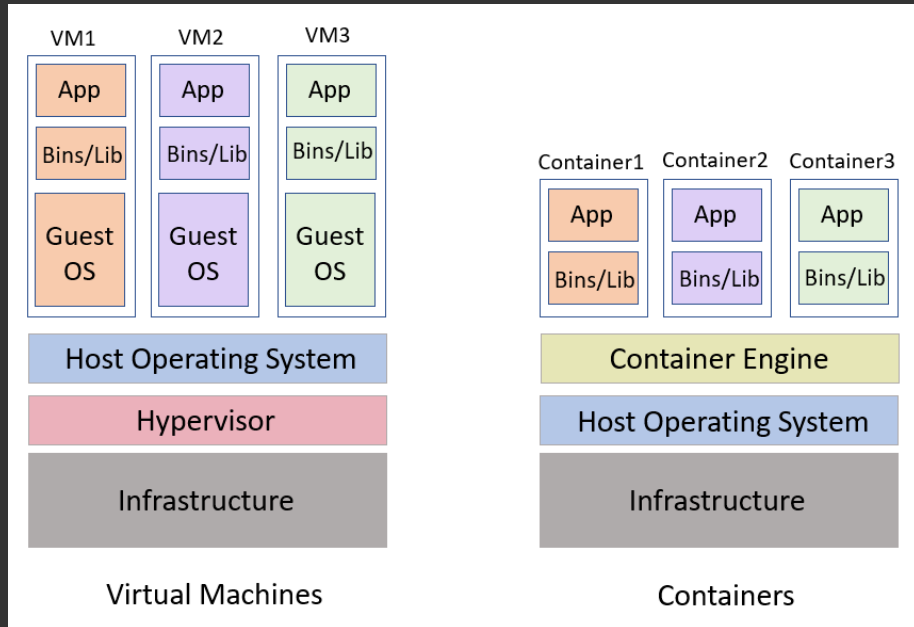




Kubernetes

Kubernetes Architecture

Docker and Docker compose



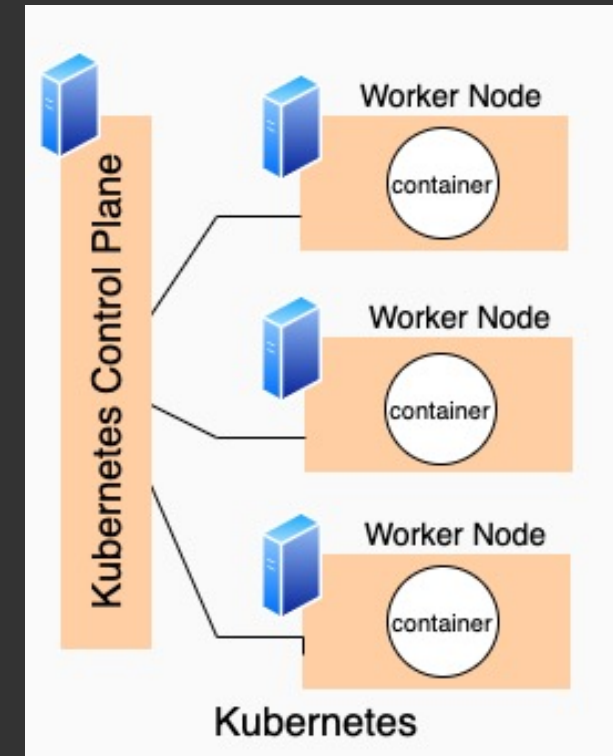
What is Kubernetes

Kubernetes is an open-source platform designed to automate the deployment, scaling, and management of containerized applications.

It allows developers to manage clusters of virtual machines (nodes) and run applications in isolated environments called containers. By organizing these containers into groups called pods, Kubernetes enables efficient resource utilization and simplifies application management.

Features of Kubernetes include:

- Self-healing
- Load balancing
- Scaling based on demand
- Helps streamline development and deployment

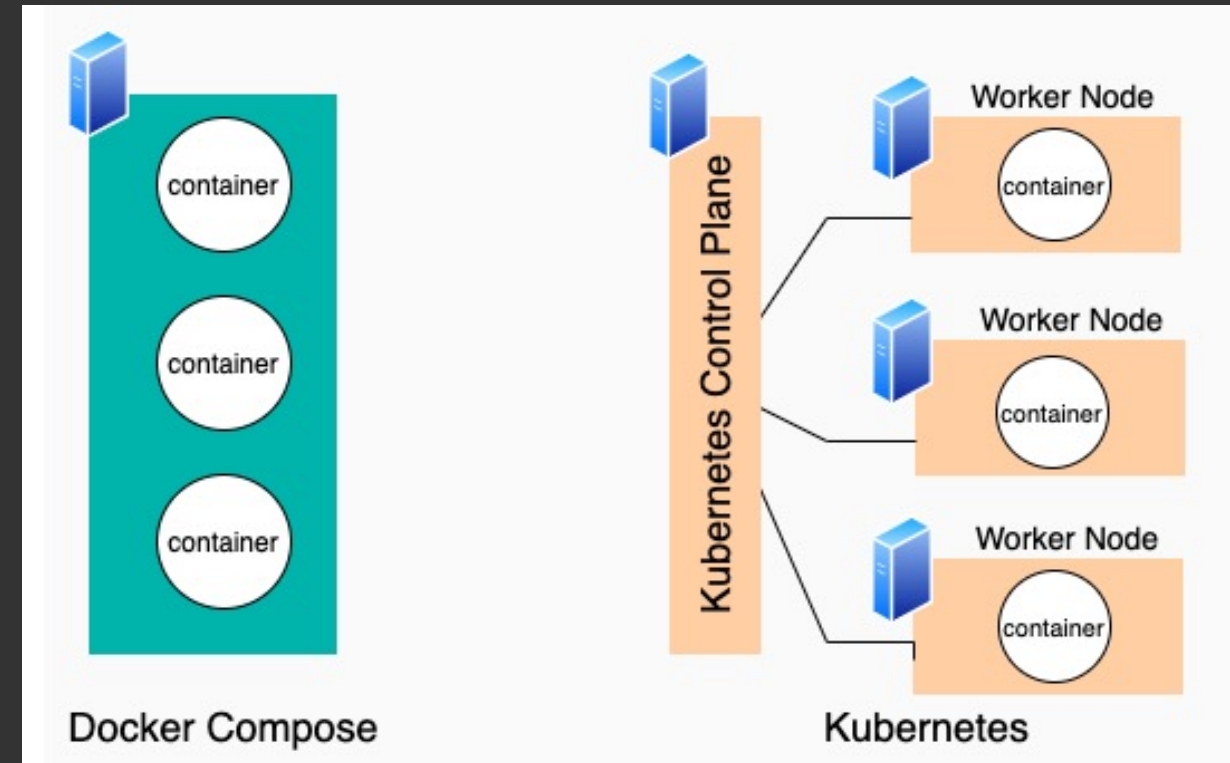


Kubernetes vs Docker compose

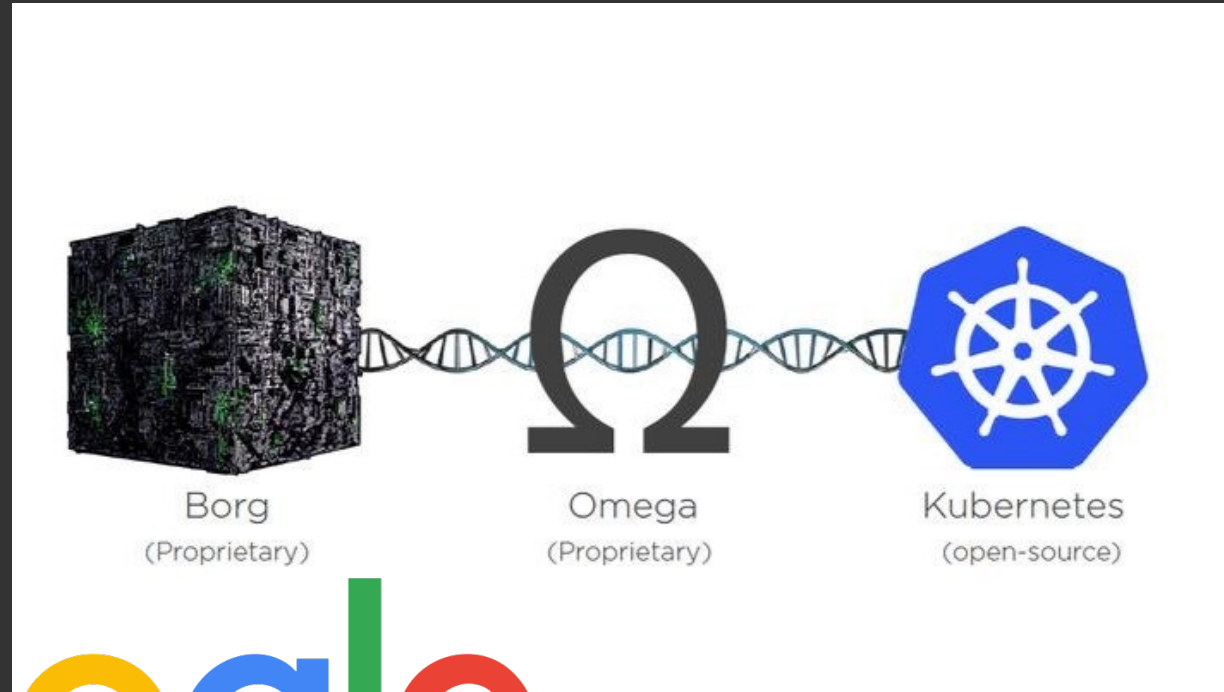
Under Docker Compose, a container and its corresponding service and network run on a single machine

while under Kubernetes containers are organized according to an abstraction called a pod which can be distributed over a number of machines.

** The Open AI Kubernetes cluster manages over 7500 Nodes [source](#)

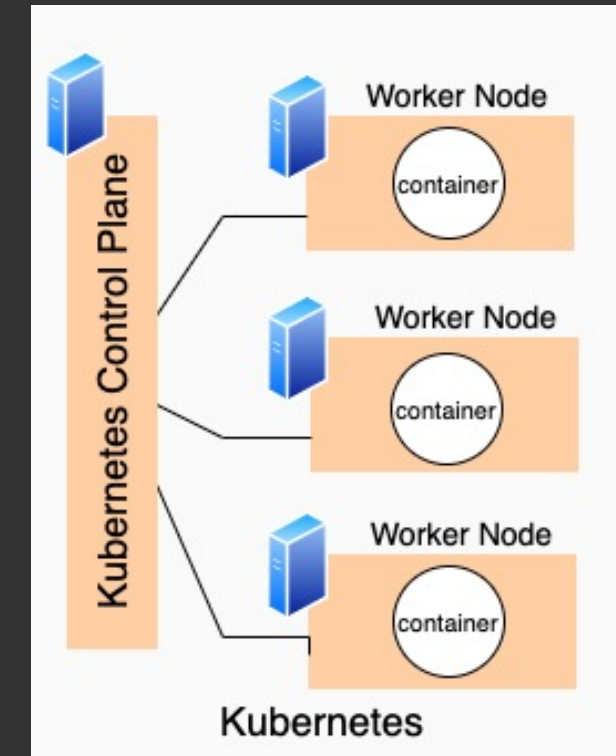
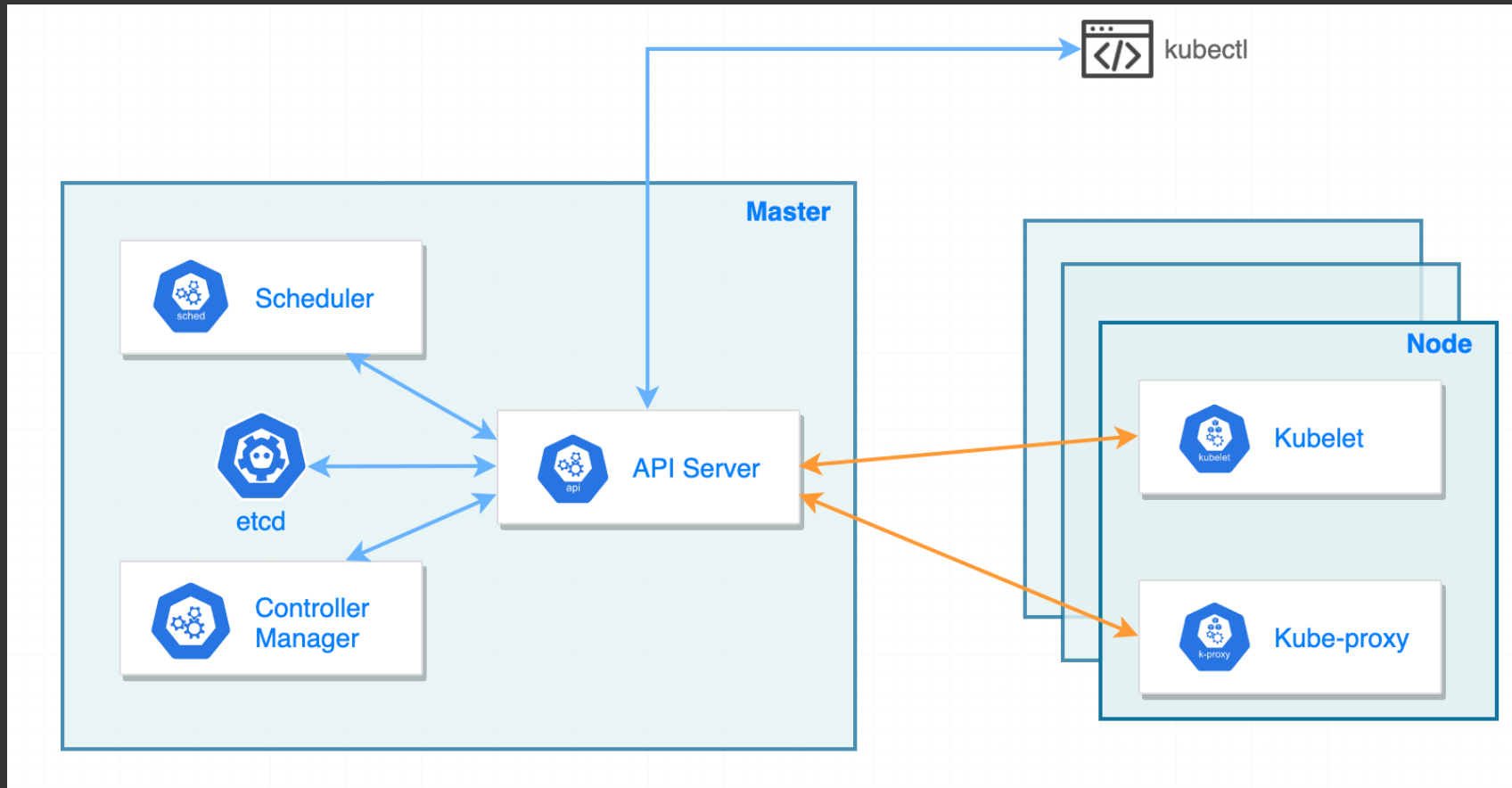


Kubernetes (k8s) Trivia



Google

Kubernetes Architecture



Kubernetes Control Plane

Control Plane: The control plane is the brain of the Kubernetes cluster, responsible for managing the cluster's overall state, making decisions about scheduling, scaling, and maintaining the desired state of applications.

- **API server**: handles requests and communication
- **etcd**: a key-value store for configuration and state
- **Controller manager**: continuously watches the state of the cluster through the API server and makes adjustments as needed to achieve the desired state.
- **Scheduler**: assigns pods to nodes based on resource availability
- **kubectl**: kubectl is the command-line interface (CLI) tool for interacting with Kubernetes clusters. It communicates with the Kubernetes API server to create, update, delete, and get information about Kubernetes resources like pods, services, deployments, and more.

Kubernetes Worker Nodes

